Davide Modolo

AI & MACHINE LEARNING ENGINEER | MSC ARTIFICIAL INTELLIGENCE

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Work Experience

OpenCity Labs - *Startup*

Remote

AI & Machine Learning ENGINEER - R&D — *August 2025 - Current*

- Engineered data pipelines processing 50k+ daily inputs for RAG ingestion, with spaCy detection and LLM fallback to fix a cross-lingual failure, reducing 35-40% language drift error; released to 10+ clients.
- Deployed a fine-tuned Transformer model for PII anonymization via NER with a reversible in-memory dictionary as a stateless, asynchronous FastAPI service on AWS EC2.
- Designed a hybrid search system combining vector embeddings with full-text reciprocal rank fusion for legal and regulatory retrieval, and built a tool-calling infrastructure with Zanzibar-based permissions via OpenFGA for user access control over tools and knowledge bases, reducing targeted operations from days to hours.
- Re-architected a product's deployment with Docker and GitLab CI/CD, automating deployment and optimizing resource allocation, reducing monthly infrastructure costs by 85% and per-instance marginal cost by 90%.

`Python, Transformers, Data Pipelines, Model Deployment, Async Programming, RAG, LLM, FastAPI, Docker, AWS EC2`

Eurecat Technology Center

Barcelona, Spain

INTERN / AI ENGINEER — *April 2024 - June 2024*

- Implemented a token-level uncertainty estimation framework for planning tasks by analyzing log-probabilities, benchmarking performance across proprietary (GPT-4o) and locally quantized models (Llama 3.1 via llama.cpp).
- Developed a RAG system utilizing *gte-large* embeddings to retrieve context-aware user preferences, significantly reducing model hallucination in recurrent decision-making scenarios.
- Prototyped a multimodal agent integrating LLaVA for visual reasoning tasks and validated the uncertainty reconstruction approach across multiple simulated environments.

`Python, PyTorch, Hugging Face, OpenAI API, llama.cpp, Ollama`

Education

University of Trento

Trento, Italy

Master's Degree in Artificial Intelligence Systems — *September 2021 - Graduated March 2025*

- **Thesis**: Exploring the Use of LLMs for Agent Planning: Strengths and Weaknesses. *[report | code]*
- **Relevant Courses**: Machine Learning, Deep Learning, Natural Language Understanding, Automated Planning, Law & Ethics in AI, Fundamentals of AI (Reasoning and Planning), Signal Processing, AI for Finance, High Performance Computing.

University of Trento

Trento, Italy

Bachelor's Degree in Computer Science — *September 2017 - Graduated June 2021*

- **Core Topics**: Algorithms & Data Structures, Software Engineering, Database Systems, Operating Systems, Computer Architectures, Distributed Networks.

Projects

COVID-19 Lung Ultrasound Images classification - Medical Imaging Diagnostic

[slides | report | code]

- Designed a multi-stage pipeline for COVID-19 LUS severity scoring (0-3): a fine-tuned ResNet18 frame classifier, a binary SVC uncertainty detector on softmax outputs, and a t-SNE-based similarity module retrieving nearest neighbours for low-confidence predictions.
- Trained and tested on 47k anonymized frames from 14 patients, with patient-level splits to prevent data leakage.

`Python, PyTorch, scikit-learn, Pandas, NumPy, Azure, Computer Vision`

Joint Intent Detection and Slot Filling - Natural Language Understanding

[report | code]

- Compared 4 architectures for joint intent detection and slot filling, from fine-tuned pre-trained transformers (BERT, ERNIE) to recurrent models built from scratch (BiLSTM, Encoder-Decoder).
- Consistently improved baseline results by at least +2% on ATIS and +13% on SNIPS datasets.

`Python, PyTorch, NLP (BERT/ERNIE/LSTM)`

Domain Adaptation / Transfer Learning - Deep Learning - *Team Project*

[code]

- Built an unsupervised domain adaptation model based on ResNet34 with a custom adaptation layer using direct 3rd-order and grouped 4th-order statistics to bridge the synthetic-to-real gap.
- Achieved +11.53% improvement over non-adapted baseline on the Adaptiope dataset.

`Python, PyTorch, Google Colab`

Skills

Languages	`Python, JavaScript, C/C++, SQL, Bash, LaTeX`
ML & Data	`PyTorch, Hugging Face, TensorFlow, Transformers, LLM, NLP, spaCy, Data Pipelines, Model Deployment, FastAPI, scikit-learn, Pandas, NumPy`
DevOps & Cloud	`Git, Docker, GitLab CI, AWS EC2, Grafana, Prometheus, MPI (High Performance Computing)`
Spoken Languages	**English** - Professional, **Italian** - Native